PH30102

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Quantum Mechanics III

Lecture 5

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Scattering and Measured Cross Sections

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$S = \Omega_-^\dagger \Omega_+$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$d\sigma/d\Omega = |f(\theta,\varphi)|^2$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$\sigma_{tot} = (4\pi/k) Im f(0)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 5; Scattering Registry SR-26; QDC acceptance appendix